

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: ARABICMMLU | **Context:** MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Content-level validity is the dimension where ArabicMMLU diverges most sharply from the deployment's eight-country requirement. While the benchmark's content is sourced from real native Arabic exams [Q1, Q26] — a clear strength over translated alternatives [Q5, Q6] — the empirical content distribution is heavily skewed: civics, social science, history, and economics at school level are dominated by Jordanian-specific institutional, royal, and constitutional content presented as universal curriculum knowledge [DATASET-D1, D2, D3, D4, D5, D6, D7, D20, D21, D22, D25, D30, D31]. University-level economics, accounting, management, and political science come exclusively from a single Egyptian university (Assiut University) in the sample [DATASET-D13, D14, D28, D39, D40], and Law (Professional) is exclusively Moroccan in the observed sample [DATASET-D8, D9, D10, D34]. The deployment user explicitly requires country-specific tailoring for law [elicitation A2] and curriculum-aligned content for each of the eight countries [elicitation A1]. Kuwait has no observed examples and no documented annotators [Q31, Q96]; Morocco has no documented annotators [Q31, Q96]. Egypt-specific currency (ج.م.ع) appears in university economics presented as universal [DATASET-D28]. The benchmark's own admission that countries are represented unequally [Q89] is empirically confirmed and amplified by the dataset analysis: rather than uniform under-coverage, the data shows Jordanian-curriculum content is propagated as default for non-Jordanian cohorts. This generates substantial construct-irrelevant variance for Moroccan, Emirati, Kuwaiti, Saudi, Lebanese, and Palestinian students.

ID	Evidence
Q1	"we present ArabicMMLU, the first multi-task language understanding benchmark for the Arabic language, sourced from school exams across diverse educational levels in different countries spanning North Africa, the Levant, and the Gulf regions." (p.1)
Q5	"Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1)
Q6	"This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)
Q26	"The questions are sourced from eight different countries in North Africa (Morocco and Egypt), the Levant (Jordan, Palestine, and Lebanon), and the Gulf (UAE, Kuwait, and KSA)." (p.3)
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q89	"ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
D1	What are the manifestations of change in modern Jordanian society? — country-specific Jordanian civics
D2	The first coalition government in Jordan in 1952 was known as the ___ government — Jordanian political history
D3	The council with general sovereignty over all state affairs is the Council of Ministers — specifically Jordanian constitutional structure
D4	Rights under the [Jordanian] Kingdom's constitution — explicitly references the Jordanian Hashemite Kingdom

D5	"لكلمة الإلحاح دعه في 1952 ماع روتسد رص" with options referring to Jordanian kings — "The 1952 constitution was issued during the reign of King [Talal bin Abdullah]" — Jordanian royal history
D6	وهو يزكركم كنبلل ينورتكلال عقومل قيرط نع يوكشل لاسرا www.cbj.gov.jo — "Complaints to the Central Bank [of Jordan] via www.cbj.gov.jo" — explicitly references Jordan's central bank
D7	0.67: نسبة من الأردنيين لا يستطيعون الوصول إلى الخدمات المالية الرسمية — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact
D8	— يوعدل رهوج في مكحل رودص دعب الإ عوفد وأ ضراع عازن في مرداصل وأ عيدي مهممتل ماكحلأ فائتتسا لبقي ال Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D9	عافدل اب فلكملم عيوكحل عطلسل او لدعل ربزو نم رارب...نيكي كدلل عيئاضلق عطرشل طباض ففص لي وخت نكمي "يكللمل كدلل" (Royal Gendarmerie of Morocco) — Moroccan judicial police powers — references
D10	— Moroccan civil party withdrawal in criminal proceedings — عحي حص عوبول عي مح: مكحل رودص لبقي ندمل فرطل لزانت اذا
D13	— عي عبتال ب عيمس نوكت لي غشتل عي ممع نإف... عيمس تل ادخل تنك اذا ونأ بساحل لوح ععجارمل بومسأ ضررتفي Computer auditing concept — general accounting principle, from Egyptian source
D14	Severely — لي ري تذت... عتفلت خسل عتيرادل إا تاي هتتدسل او صاختل ألأ ني تب عطلتدل عتيري هت عتجرد OCR-distorted text about authority distribution in organizations, from Egyptian source
D20	"Royal Medical Services include: Medical City Hospital" — عي بطلل عني دمل ي فشتسم: اهنم... تاسس ؤمل نم دي دعل عيوكحل عيكللمل عي بطلل تامدمل ل مشرت specifically Jordanian Royal Medical Services
D21	"King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — عنس همك ح دموم، 1952: في كل ذو عي حص بابسأل هتالو تهتتا نيسحل نب لال طلكلمل Jordanian royal history in primary school
D22	— Jordanian religious geography — "Lady of the Mountain Church is in Anjara [Jordan]" — عرجنع في لبجلل عديس عسي نك دجوت
D25	"The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — عبادتنال ك ف دويق نم دالبل رحت: ١٩٢٨ ماع هي ناطي ربل هي ندرألأ هدا عملل سيئرل فدهل ناك Jordanian national history
D28	— Egyptian currency (جها = ١٠٠٠٠٠ بولطم رامثتسال مجح ناك اذا) ١٠ %: وه عرمثتسمل ل او مالل عل عئاعل لدعم... هي نك ١٠٠٠٠٠ بولطم رامثتسال مجح ناك اذا Egypt-specific monetary context
D30	"Wadi Rum is a Jordanian nature reserve under joint management" — مري داو... كعرتشم عرادل اهيل عل فرشت ندرألأ في عي عي بط عي محم Jordan-specific geography fact
D31	King Abdullah I's positions on Zionism — عي طسلف في اعامطأو عي نو يهصلل كرحل نم لوال هل لادبع لكلمل فق او نم تسيل عي تال نم دحاو highly sensitive political content, Jordanian framing of Palestinian history
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
D39	Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source
D40	Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source

Strengths

- Native exam provenance avoids translation artifacts that plague competing Arabic benchmarks [Q5, Q6]
- STEM content (math, biology, physics, CS factual) is genuinely curriculum-neutral and reliable across all eight countries [DATASET-D24]
- Palestine is substantively represented in STEM subjects (Biology HS, CS Primary, Math Primary) [DATASET-D23, D24]
- Some Moroccan legal content is present, even if narrow [DATASET-D8, D9, D10]
- Native annotators and exam URLs document content provenance per question [Q32, Q33]

ID	Evidence
Q5	"Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1)
Q6	"This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)

Q32	"During the first stage of data collection, the internal workers were tasked with collecting relevant sources for data collection. These sources were URLs containing the questions, which needed to be publicly available." (p.4)
Q33	"In the second stage, all workers were asked to manually scrape the data within a 2-month period. The task was to collect metadata, including the source (URL of the source document), country, subject, level, question, multiple-choice options, and the correct answer key." (p.4)
D8	— Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure يؤعدل رهوج يف مكحل رودص دعب الإ عوفد وأ ضراع عازن يف ةرداصل وأ أيديهم التامك حأل افانئتسا لبقي ال
D9	عافدل اب ةفل كم لم ةيموكحل اطلسل او لدعل ربزو نم رارقب...نني كيردل ل ةيئاضق ل طرشل طباض ةفص لي وخت نكمي Moroccan judicial police powers — references "لكم ل كيردل" (Royal Gendarmerie of Morocco)
D10	Moroccan civil party withdrawal in criminal proceedings — عحي حص ةب و جأل عي ج : مكحل رودص لبقي ندمل فرطل ل زانت اذا
D23	"The binary number system is used by computers" — Palestine-sourced primary CS دأ
D24	Calvin cycle question — ؟ اهتيبثت مت يتي الت (CO2) تائي زج ددع مكف نفل اك قحل نع زوكولغ تائي زج (3) جتن اذا Palestine-sourced biology; universal factual content

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Inputs heavily require region-specific knowledge but the regions covered diverge from the deployment's eight-country target. Civics and history items reference Jordanian royal lineage, the 1952 Jordanian constitution, Jordan's Royal Medical Services, and Central Bank of Jordan URLs as universal facts [DATASET-D1, D2, D4, D5, D6, D20, D21, D25, D31]. University accounting and economics use Egyptian institutional and currency context [DATASET-D28, D39, D40]. Law uses Moroccan procedural code [DATASET-D8, D9, D10].

IC-2: Cultural-content alignment is partial. Islamic studies content observed (Hadith ethics, Quran recitation etiquette) appears broadly Sunni-consistent, aligning with the deployment's predominantly Sunni population [DATASET-D17, D37, D38]. However, no madhab calibration exists [gap_id 7], and Hanbali/Maliki specificity is not verified. Politically sensitive content (e.g., King Abdullah I and Zionism [DATASET-D31]) is framed in Jordanian national narrative — potentially misaligned with Palestinian, Lebanese, or other framings.

IC-3: Western-specific knowledge is largely avoided due to native exam sourcing [Q5, Q6]. The primary content-validity threat is not Western bias but intra-regional bias — Jordanian and Egyptian content dominating non-Jordanian/non-Egyptian cohorts.

IC-4: Annotators are exclusively from Jordan, Egypt, Lebanon, UAE, KSA [Q31] — Morocco, Kuwait, Palestine have no documented annotators despite (in Palestine's case) contributing top-three question volume [Q96, WEB-1]. Recruiting Kuwaiti, Moroccan, and Palestinian curriculum experts would be essential for any deployment use.

IC-5: Documented content issues harming validity: (a) Jordanian-default civics/history/social science presented as universal [DATASET-D1, D2, D4, D21]; (b) single-Egyptian-university dominance of university social science subjects [DATASET-D39, D40]; (c) single-source single-jurisdiction Law (Professional) [DATASET-D34]; (d) no Kuwait content observed; (e) Egypt-specific currency in 'general' economics [DATASET-D28]; (f) French-medium Moroccan university content absent [Q92].

ID	Evidence
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Q5	"Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1)
Q6	"This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q89	"ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)
Q92	"The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
D1	"What are the manifestations of change in modern Jordanian society?" — country-specific Jordanian civics
D2	"The first coalition government in Jordan in 1952 was known as the ___ government" — Jordanian political history
D4	"Rights under the [Jordanian] Kingdom's constitution" — explicitly references the Jordanian Hashemite Kingdom
D5	"The 1952 constitution was issued during the reign of King [Talal bin Abdullah]" — Jordanian royal history
D6	www.cbj.gov.jo — "Complaints to the Central Bank [of Jordan] via www.cbj.gov.jo" — explicitly references Jordan's central bank
D8	appeals rules under Moroccan Code of Civil Procedure
D9	references "Royal Gendarmerie of Morocco" (Moroccan judicial police powers)
D10	Moroccan civil party withdrawal in criminal proceedings
D17	Country: Jordan — 5-option Islamic studies question about positive effects of thinking
D20	"Royal Medical Services include: Medical City Hospital" — specifically Jordanian Royal Medical Services
D21	"King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school
D25	"The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — Jordanian national history
D28	Egyptian currency (= Egyptian pound) in university economics — Egypt-specific monetary context
D31	King Abdullah I's positions on Zionism — highly sensitive political content, Jordanian framing of Palestinian history
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
D37	Country: Jordan — Generic Islamic etiquette of Quran recitation — no madhab specificity indicated
D38	Country: Jordan — Islamic studies question about examples in Quran — Jordanian curriculum framing
D39	Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source
D40	Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source
WEB-1	Models perform notably poorly on Morocco- and UAE-sourced questions, consistent with thin annotation coverage (https://arxiv.org/html/2402.12840v1)

Information Gaps

- Whether Jordanian dominance in civics/history/social science extends to the full benchmark or is concentrated in specific level/subject configurations [gap_id 5]
- Whether Law subset beyond the sampled Moroccan questions includes other jurisdictions [gap_id 4]
- Madhab-specific framing in Islamic studies is undocumented [gap_id 7]

Requires Expert Verification

- Per-country curriculum review by national-curriculum experts from Kuwait, Morocco, Palestine, UAE to identify questions presented as 'universal' that are actually national-curriculum-specific
- Islamic studies curriculum experts to assess madhab calibration risk for Moroccan Maliki and Saudi Hanbali contexts

Remediation

Gap 1: Jordanian-curriculum civics, history, social science, and high-school economics content is encoded as universal across configurations, creating construct-irrelevant variance for the seven non-Jordanian deployment cohorts [DATASET-D1, D2, D6, D7, D21, D25].

Recommendation: Use the per-question country metadata field [Q33] to filter or stratify evaluation by source country; do not aggregate across countries for civics, history, or social science. Build a Kuwaiti, Moroccan, and Palestinian curriculum supplement before deploying to those cohorts.

Gap 2: Islamic studies has no madhab labeling anywhere in the benchmark [gap_id 7]; observed Islamic Studies HS content is Jordanian-curriculum-sourced [DATASET-D38] without Maliki (Morocco) or Hanbali (KSA) calibration.

Recommendation: Engage Islamic studies curriculum experts from Morocco (Maliki), KSA (Hanbali), and a Shafi'i-region representative to audit a stratified sample of Islamic studies items and tag madhab-specific items; flag madhab-divergent questions for country-conditioned evaluation.

ID	Evidence
Q33	"In the second stage, all workers were asked to manually scrape the data within a 2-month period. The task was to collect metadata, including the source (URL of the source document), country, subject, level, question, multiple-choice options, and the correct answer key." (p.4)
D1	"What are the manifestations of change in modern Jordanian society?" — country-specific Jordanian civics
D2	"The first coalition government in Jordan in 1952 was known as the ___ government" — Jordanian political history
D6	www.cbj.gov.jo — "Complaints to the Central Bank [of Jordan] via www.cbj.gov.jo" — explicitly references Jordan's central bank
D7	0.67 — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact
D21	"King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school
D25	"The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — Jordanian national history
D38	"Country: Jordan — Islamic studies question about examples in Quran — Jordanian curriculum framing